

International Islamic University on Chittagong
Center for General Education (CGED)
Midterm Examination-Spring-2026

Course Title: Life and Teachings of Prophet (SAAS) Course Code: URED-3604
(URED-3201 for LLB)

Time 1:30 Hours

Full Marks: 30

Answer all questions strictly

#	Questions	Marks	CLOs	Bloom's taxonomy domain
1	A) What do you understand by <i>Sirah</i>? Discuss the benefit of studying the <i>Sirah</i> of the Prophet (SAAS) in the light of the Qur'an and <i>Sunnah</i>. Or, B) What is <i>Ayyam Jahiliyyah</i>? Give a brief socio-political background of <i>Ayyam Jahiliyyah</i>.	10	2	Remember & Understand
2	Why the disbelievers of Makkah turned against the Prophet Muhammad (SAAS) despite their testimony to him as <i>al-Ameen</i> (most trustworthy) and <i>al-Sadique</i> (most truthful)? Describe the attempts taken by the Quraysh to stop him from his missions?	10	2	Understand
3	Write short notes on the following topics: (Any two) a. <i>Hilful Fudul</i> b. Boycott and confinement c. Journey to <i>Ta'if</i>	10	2	Remember

International Islamic University Chittagong

Center for General Education (CGED)

Midterm Examination: Spring 2026

Course Code: GEHE-3601

Program: Undergraduate

Course Title: History of Emergence of Bangladesh

Time: 1 hours and 30 minutes.

Full Marks: 30

Instructions:

- All Questions are Compulsory.
- Figures in the right margin indicate full marks.
- Course Learning Outcome (CLO) and Bloom's levels are mentioned in additional columns.

Bloom's Levels of the Questions.

Letter of Symbol	R	U	App	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

	Text of the Questions	Marks	Bloom's Level	CLO
1	Critically explain the area and boundaries of Bangladesh and sketch the major physiographic units of the country. Or Investigate the ethnicity of the people of Bangladesh. Why is the population of Bangladesh considered ethnically heterogeneous?	10	An	CLO1
2	Explain the Socio-political context of the 'United Independent Bengal' movement in 1947. What were the key factors behind its ultimate failure?	10	An	CLO2
3	Review the historical background of the Language Movement of 1952 and evaluate its significances in the development of Bengali nationalism.	10	E	CLO2

International Islamic University Chittagong
Department of Computer Science & Engineering

Program: B. Sc. in CSE

Mid-Term Examination, Spring-2026

Course Code: ECON-3501

Course Title: Principles of Economics

Time: 1 Hour 30 Minutes

Full Marks: 30

- (i) Answer all the questions. The figures in the right-hand margin indicate full marks.
(ii) Course Learning Outcomes (COs) and Bloom's Levels are mentioned in additional columns.

Course Learning Outcomes (CLOs) of the Questions

CLO1	Demonstrate knowledge of the fundamental concepts and theories of micro and macro-economics.
CLO2	Identify the key indicators of economic growth and how to use the indicators for national and international comparisons.
CLO3	Identify and explain the key macroeconomic policies.
CLO4	Use economic theories and concepts to analyze behavior of individuals, firms and nations to act as a responsible citizen.

Bloom's Level of the Questions

Letter Symbols	R	U	A	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

Part-A: Answer the following questions

Q1	a	What are the basic economic problems?	CO1	U	2
	b	Explain how unlimited wants and limited resources create the fundamental economic problem.	CO1	R	4
	c	Evaluate the free market hypothesis. Can markets always reach equilibrium?	CO1	R	4
OR					
	a	Define the Production Possibility Curve (PPC) and explain how it demonstrates the concept of opportunity cost. Illustrate with examples why it is usually concave.	CO1	U	5
	b	How would you describe the context scarcity and choice addressing the central problems of economics?	CO1	R	5
Q2	a	The demand and supply functions for a commodity are given as: $Q_d = 20 - 4P$ $Q_s = 8 + 3P$ I. Construct the Demand and Supply Schedule for prices $P = 3, 5, 7, 9, 10$. II. Plot the demand and supply curves using these schedules. III. Explain how your results illustrate the Laws of Demand and Supply. IV. Determine the equilibrium price and equilibrium quantity.	CO2	An	6
	b	Due to the US-Iran conflict, the price of petrol rises from Tk. 100 to Tk. 200 per litre, and weekly motorcycle sales fall from 500 to 300 units. Identify the type of elasticity shown and explain why.	CO2	E	4
Q3	a	"Marginal productivity determines factor payment" discuss the statement with example.	CO2	E	3
	b	Define the indifference curve and how does it relate with the concept of marginal utility.	CO2	An	4
	C	What is returns to scale? Discuss the various types of returns to scale.	CO2	E	3

International Islamic University Chittagong

Department of Computer Science & Engineering

Program: B. Sc (Engg.) in CSE Semester: 6th

Mid-Term Examination, Spring 2026

Course Code: CSE-3641

Course Title: Software Engineering

Time: 1 hour and 30 minutes

Total Marks: 30

[Answer all the questions. Figures in the right-hand margin indicate full marks.]

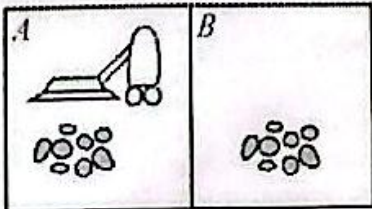
	M	CLO	PLO	DL
Q.1. a. A project team develops an initial version of an e-commerce website and shares it with customers to <u>gather feedback</u> . Based on customer suggestions, the team refines and <u>updates the system</u> before moving to the final implementation.	5	CLO1	PLO1	C3
1. Which process model is being used here?				
2. What advantages and disadvantages does this model bring to this case?				
Or				
A company is tasked with building software for a life-support device that must work without failure under strict regulations. Since the project involves high risk,	5	CLO1	PLO1	C3
1. Which software development model would be most appropriate in this case?				
2. Justify your choice with respect to handling risks and ensuring thorough validation.				
b. A company follows a traditional development approach where developers and operations teams work separately. Software updates are deployed <u>manually</u> , leading to frequent errors, system downtime, and <u>delayed releases</u> . There is <u>no automated</u> testing or monitoring system in place, and communication between teams is poor.	5	CLO1	PLO1	C3
1. What are the key problems in this scenario?				
2. How can you resolve these issues?				
Q.2. Read the passage and answer questions a, b, and c: MediCare Solutions is building a hospital management system that covers patient registration, doctor scheduling, billing, and pharmacy management. All requirements have been clearly defined upfront by hospital administration, the scope is fixed, and the project has a strict regulatory approval process that requires complete documentation at every phase. A junior developer suggests using the Prototyping model to get faster user feedback. A senior engineer disagrees and recommends Waterfall.				
a. The junior developer's suggestion of Prototyping is well-intentioned but unsuitable here. Explain why it is a poor fit for this specific project.	3	CLO1	PLO1	C3
b. Justify the senior engineer's recommendation of the <i>Waterfall model</i> by mapping at least <i>three</i> of the project's characteristics to the specific strengths of Waterfall. Also mention <i>one</i> limitation of the waterfall that the team should still be cautious about.	4	CLO1	PLO1	C4
c. List the six phases of the Waterfall model in correct sequential order and briefly describe what activity takes place in each phase as it would apply to the MediCare hospital system.	3	CLO1	PLO1	C2

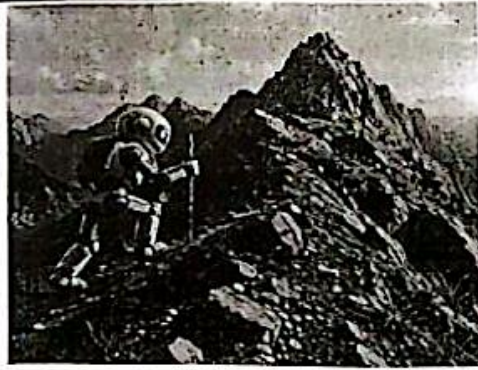
Or

- Q.2.** A defense contractor is developing flight control software for a military drone. The project involves highly uncertain technical requirements, significant risk of catastrophic failure, and demands continuous review and improvement at every stage. The development team is evaluating three models: Waterfall, Incremental, and Spiral.
- Which model is the most appropriate choice for this project? Your answer must reference all phases of the selected model and explain what happens in each phase in the context of this drone project. 5 CLO1 PLO1 C3
 - Compare the Spiral and Incremental models in terms of how each handles project risk. Use the drone project to illustrate the difference. Which model better handles a situation where a critical flaw is discovered late? 3 CLO1 PLO1 C4
 - The project manager wants to skip the Risk Analysis to save time. As the lead software engineer, how do you respond? Refer to the costs of software engineering to support your argument. 2 CLO1 PLO1 C5
- Q.3**
- A technology company is developing a cloud-based Software-as-a-Service (SaaS) platform that provides data analytics solutions to multiple clients. The system must support thousands of concurrent users and ensure high availability and scalability. It uses a microservices architecture where each service is independently deployable. The development team follows DevOps practices and relies on automated CI/CD pipelines to ensure continuous integration, testing, and deployment.
 - Identify key non-functional requirements.
 - Why is DevOps critical in this system? How does CI/CD pipeline support scalability and reliability?
 - A fintech application is being developed to handle sensitive financial data and must comply with strict regulatory requirements imposed by central banks and international data protection laws. The system must ensure data security, privacy, and legal compliance while providing seamless services to users. Failure to meet these requirements could lead to legal penalties and loss of customer trust. How do these requirements affect system design? What are the risks of ignoring regulatory requirements? 5 CLO2 PLO4 C5

Bismillahir Rahmanir Rahim
International Islamic University Chittagong
Department of Computer Science & Engineering
B. Sc. in CSE Semester Midterm Examination, Spring-2026
Course Code: CSE-3635 Course Title: Artificial Intelligence
Total marks: 30 Time: 1 hour 30 minutes

[Answer *all* the questions. Figures in the right-hand margin indicate full marks]

			CLO	Marks
1	a)	Define in your own words for an AI agent: i. Percept Sequence ii. Episodic Environment	CLO1	2
1	b)	Do you think AI will create more jobs than it destroys, or the opposite? Give two arguments to support your answer.	CLO1	2
1	c)	Consider an Autonomous Campus Mail Carrier. This agent is designed to navigate a university campus autonomously to deliver documents and small packages between different department buildings while avoiding pedestrians, cyclists, and campus shuttle buses. a) Define the PAGE (Percepts, Actions, Goals and Environment) specification for this autonomous agent. b) Identify and justify the environment type for this agent by selecting the correct category for the following dimensions: i. Fully Observable vs. Partially Observable ii. Deterministic vs. Stochastic c) Identify whether this agent should be a Simple Reflex Agent or a Goal-Based Agent.	CLO1	2+2+2
		OR		
1	c)	Develop a state space diagram and a partial tabulation representation containing percept sequences and actions for a vacuum-cleaner world shown in Figure 01- 	CLO1	3+3
2		An autonomous robot is deployed in a mountainous region to reach the highest peak. The environment contains multiple peaks, flat regions, and narrow ridges.		



Initially, the robot uses a greedy strategy where it always moves to the best neighboring state. However, it frequently gets stuck before reaching the highest peak.

To overcome this, engineers upgrade the system using a probabilistic approach that occasionally allows worse moves, especially at the beginning.

2	a)	Analyze why the robot fails using Hill Climbing. Explain with related graph for each of the following cases: • Local maxima • Plateau • Ridge	CLO2	2+2+2
2	b)	Explain how Simulated Annealing overcomes these limitations.	CLO2	2
		Or		
2	b)	Explain the role of the temperature parameter in Simulated Annealing and how it changes the robot's behavior over time.	CLO2	2
2	c)	Why is Hill Climbing called a "greedy local search algorithm"?	CLO2	2
3	a)	What is forward checking in constraint satisfaction problems? How does it help in solving problems efficiently? Explain with respect to the map coloring problem.	CLO2	2
3	b)	Suppose, the initial population of an 8-queen problem consist of four individuals as follows: X1=24748552, X2=32752411, X3=24415124, X4=32543213 Let the fitness of individual X be equal to the number of non-attacking pair and the score of X1 and X2 is 24 and 23 respectively. i) Evaluate the fitness of X3 and X4, showing all your calculations. ii) Perform crossover operation on the fittest two individuals at the third point. iii) Perform multi-gene mutation on the generated individual from (ii) and show the result.	CLO2	4+2+2

(i) The figures in the right-hand margin indicate full marks

(ii) Course Outcomes and Bloom's Levels are mentioned in additional Columns

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|----|------|--|-----|----|---|
| 1 | a) | Define data communications and mention its components. Briefly explain simplex, half duplex and full duplex communication with example. | CO1 | U | 5 |
| | b) | What do you mean by protocol and why protocol layering is important? Also, for each layer of TCP/IP model mention the data unit or PDU name. | CO1 | U | 5 |
| 2 | a) | Define attenuation, distortion and noise. What is SNR and SNR _{db} . | CO1 | U | 5 |
| 2. | b) | Find SNR _{DB} for noiseless channel using appropriate equation (consider it as an ideal, not for real life)? Assume that SNR _{DB} = 36 and Channel bandwidth is 1 Mhz. Find the appropriate bit rate and signal level? | CO2 | An | 5 |
| OR | | | | | |
| 2 | b) | As a telecom engineer, you need to send 10 Gbps data over 10 km using optical fiber. Should you choose <u>single-mode</u> or multi-mode fiber? Explain your choice with respect to propagation modes and modal dispersion. | CO2 | An | 5 |
| 3. | a) | Draw digital signal for the following line coding schemes using the bit stream <u>11000001</u> : | CO3 | An | 4 |
| | i. | NRZ-I | | | |
| | ii. | Manchester | | | |
| | iii. | Differential Manchester | | | |
| | iv. | Pseudoternary | | | |
| | v. | AMI with HDB3 | | | |
| 3. | b) | Carry out PCM for an sinusoidal analog signal that has highest amplitude of +16V and lowest amplitude of -16 V and assume you are using 4 bits to represent each sample. | CO3 | An | 5 |
| OR | | | | | |
| 3 | b) | We want to digitize a signal, which ranges between 0 to 6000Hz. Find the sampling rate and bit rate if we use 4 bits per sample. Also find the ratio of minimum bandwidth of digital signal to bandwidth of the analog signal. | CO3 | An | 5 |



International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE

Mid Exam, Spring-2026

Course Code: CSE 3631

Course Title: Operating System

Time: 1 hours 30 minutes

Full Marks: 30

- (i) The figures in the right-hand margin indicate full marks
(ii) Course Outcomes are mentioned in additional Columns

[Answer the questions from the followings]

1. a) List four services provided by operating system and explain how each creates convenience for users. Explain the difference between program and process. C01 05
1. b) Define system calls and explain their significance in an operating system. Discuss in detail how system calls act as an interface between user-level programs and the kernel. Illustrate the complete execution process with a suitable example. C01 05

OR

1. b) Describe the characteristics of a batch operating system in detail. Explain how jobs are processed in batches without user interaction and analyze its advantages and disadvantages compared to modern interactive operating systems.
2. a) A cloud computing platform is processing multiple tasks with varying burst times. The server uses Shortest Remaining Time First (SRTF) scheduling to ensure that tasks with the shortest remaining burst time are processed first. The tasks are submitted at different times, and the platform needs to decide how to schedule them. Task Arrival Time (ms) Burst Time (ms). Once selected, each process is given the CPU for a fixed time quantum of 2 units. C02 05

Process	Arrival Time	Burst Time	Priority
P1	3	1	2
P2	1	4	1
P3	4	2	3
P4	0	6	2
P5	2	3	5

Simulate the execution of these tasks using SRTF scheduling.

- i) Draw the Gantt chart.
- ii) Calculate the waiting time and turnaround time for each task.

2. b) Consider a set of processes P1, P2, P3 and P4 with the following arrival times, burst times, and priorities: C02 05

Process	Arrival Time	Burst Time	Priority
P1	0	10	2
P2	1	15	1
P3	2	13	3
P4	3	11	2

The system uses a hybrid scheduling (RR+Priority) approach where processes are selected based on Priority Scheduling. Once selected, each process is given the CPU for a fixed time quantum of 5 units. After using its time quantum, if the process is not finished, it is placed back in the ready queue based on priority.

- i) Draw the Gantt chart for the execution of processes.
- ii) Find the average waiting time.

3. a) Semaphores are an essential synchronization tool used by operating systems to solve the critical section problem and manage concurrent access to shared resources. Describe the structure and operation of semaphores and evaluate their strengths and weaknesses as a synchronization mechanism. CO3 05

3. b) Explain the structure and working mechanism of a mutex lock with suitable diagrams or pseudocode. Discuss how locking and unlocking operations are implemented in an operating system. CO3 05

OR

3. b) A drone system schedules the following periodic tasks using Earliest Deadline First (EDF):

T1 (Obstacle Detection): Execution = 2 ms, Period = 6 ms, Deadline = 6 ms

T2 (Flight Stabilization): Execution = 1 ms, Period = 4 ms, Deadline = 4 ms

T3 (Video Streaming): Execution = 3 ms, Period = 12 ms, Deadline = 10 ms

Tasks:

- i) Identify if any deadlines are missed under normal conditions.
- ii) Suggest a modification to improve performance.